TierZero

Autonomous AI Ticket Resolution

Pick up a ticket. Understand the problem. Search knowledge. Do the work. Close the ticket.

No human intervention. No brittle scripts. Fully autonomous.

Steve Elliott -- March 2026

What TierZero Does

An AI agent that **autonomously resolves IT support tickets** -- from reading the ticket to closing it -- without human involvement.

🎫 Picks Up Tickets

Polls ServiceNow, Jira, or GitLab for new open tickets. Reads the full description and comment thread.

⚡ Executes the Fix

Browser automation, code changes, or guided responses. Adapts to UI changes. Self-heals when things break.

🧠 Understands the Problem

LLM analyzes the ticket against ingested knowledge base. Determines what kind of work is needed.

✅ Closes the Ticket

Posts resolution, audit trail, KB sources cited. Marks ticket resolved. Moves to the next one.

🔍 Searches for Solutions

RAG retrieval across runbooks, wiki pages, past tickets, and documentation. Finds the procedure.

🚨 Knows When to Escalate

Security incidents, low confidence, policy decisions -- TierZero escalates with full context instead of guessing.

The Architecture

Five-node LangGraph pipeline. Every ticket flows through the same graph:

📥 Ingest
Load ticket + comments
→
🔍 Retrieve
RAG search KB
→
🧠 Decide
LLM structured output
→
⚡ Act
Execute decision
→
📝 Record
Audit trail

Decide Node Output (5 decisions):

- **automate** -- Full resolution exists in KB. Execute and close.
- **draft_response** -- Guidance found. Post helpful reply.
- **escalate** -- Out of scope or low confidence. Hand to human with context.
- **needs_info** -- Ticket too vague. Ask one specific question.
- **implement** -- Bug/feature. Write code, create branch, run tests.

Safety Rules (built into the LLM prompt):

- ALWAYS escalate security incidents
- ALWAYS escalate production system changes
- Never repeat a fix the user already tried
- Confidence below 0.4 = auto-escalate
- Cite KB sources in every response

Knowledge Ingestion

TierZero learns from your existing documentation. No training required -- just point it at your knowledge sources.

📚 Sources Supported

- **Azure DevOps Wiki** -- full wiki + work items
- **Confluence** -- spaces, pages, attachments
- **URLs** -- scrape any documentation site
- **Past Tickets** -- mine ServiceNow/Jira/GitLab for resolved tickets with solutions
- **Local docs** -- any folder of markdown/text/PDF files

🔧 How It Works

- `tierzero index ./knowledge` -- chunk and embed into ChromaDB
- `tierzero import-wiki --source azuredevops` -- pull from AzDO
- `tierzero mine-tickets --connector servicenow` -- learn from past tickets
- `tierzero import-url https://docs.example.com` -- scrape web docs
- Incremental updates -- only re-indexes changed files

Key: The RAG pipeline uses MMR (Maximal Marginal Relevance) for diverse results -- so it doesn't just return 5 copies of the same runbook.

Autonomous Execution: The Watch Loop

TierZero runs continuously, polling for new tickets and resolving them without human intervention.

```
$ tierzero watch --instance-url https://company.service-now.com --interval 60

[14:32:01] Watching for open tickets interval: 60s
[14:32:01] → INC0401474 VPN not connecting from remote office
           ✓ automate (0.91) action: resolved
[14:32:08] → INC0401489 Suspicious login from unknown IP
           ! escalate (0.95) action: escalated to Security
[14:33:01] → INC0401502 Can't print to 3rd floor printer
           ✓ automate (0.87) action: resolved
[14:33:15] → INC0401510 Need access to shared drive
           ? needs_info (0.72) action: requested_info
[14:34:01] No new open tickets
```

Your queue is zero by morning. That's the pitch. Not "here's a tool" -- "here's an outcome."

Code Implementation

When the ticket is a bug or feature request, TierZero doesn't just triage -- it **writes the code**.

What the agent does:

- Creates a feature branch (`tierzero/INC-1234`)
- Analyzes the codebase for relevant files
- Writes the fix using Claude/GPT/Gemini
- Runs the test suite
- Commits with descriptive message
- Posts the diff summary to the ticket
- Supports OpenAI, Anthropic, Google models

```
Agent output:
✓ Implemented on branch tierzero/INC-1234
  Commit: a2f8c91
  Files: 3 changed
  Tests: passing

Ticket comment:
Code changes on branch tierzero/INC-1234.
Fixed null reference in CoveragesMapper
when cancel quotes return null coverages.
3 file(s) changed. All tests passing.
```

Self-Healing Browser Automation

When the work requires browser interaction (ServiceNow forms, admin panels, portals), TierZero uses **intent-based automation** that adapts to UI changes.

The Problem with Scripts:

```
page.click('#btn-resolve')
```

UI redesign → selector breaks → script dies

Button renamed → text match fails → script dies

New modal appears → script dies

TierZero's Approach:

```
goal: "Click the Resolve button"
```

UI redesign → finds element by accessibility role

Button renamed → LLM understands context

New modal → auto-dismissed by recovery system

5-Strategy Fallback Chain:

- **1. Cached** (~50ms) -- last-known selector
- **2. Aria** (~200ms) -- accessibility role/label
- **3. LLM/A11y Tree** (~500ms) -- GPT-4o-mini
- **4. LLM/Vision** (~1s) -- screenshot analysis
- **5. Coordinates** (~1.5s) -- pixel-level click

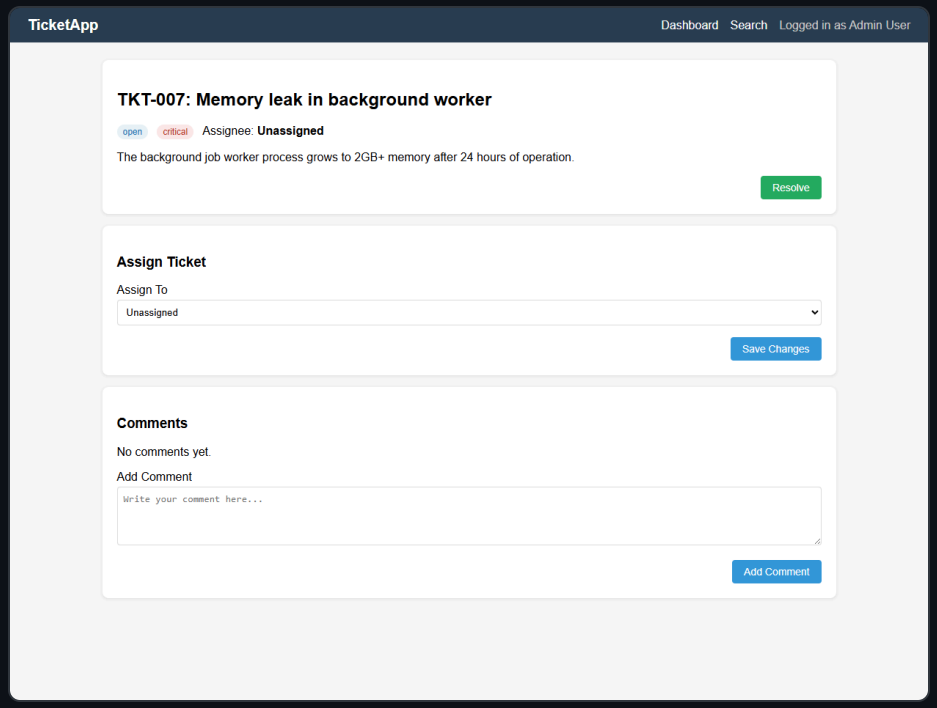
Recovery Strategies:

- Auto-dismiss modals/dialogs
- LLM page state analysis
- Navigate away from error pages
- Escalate with explanation if all else fails

Proof: Record on v1, Replay on v2

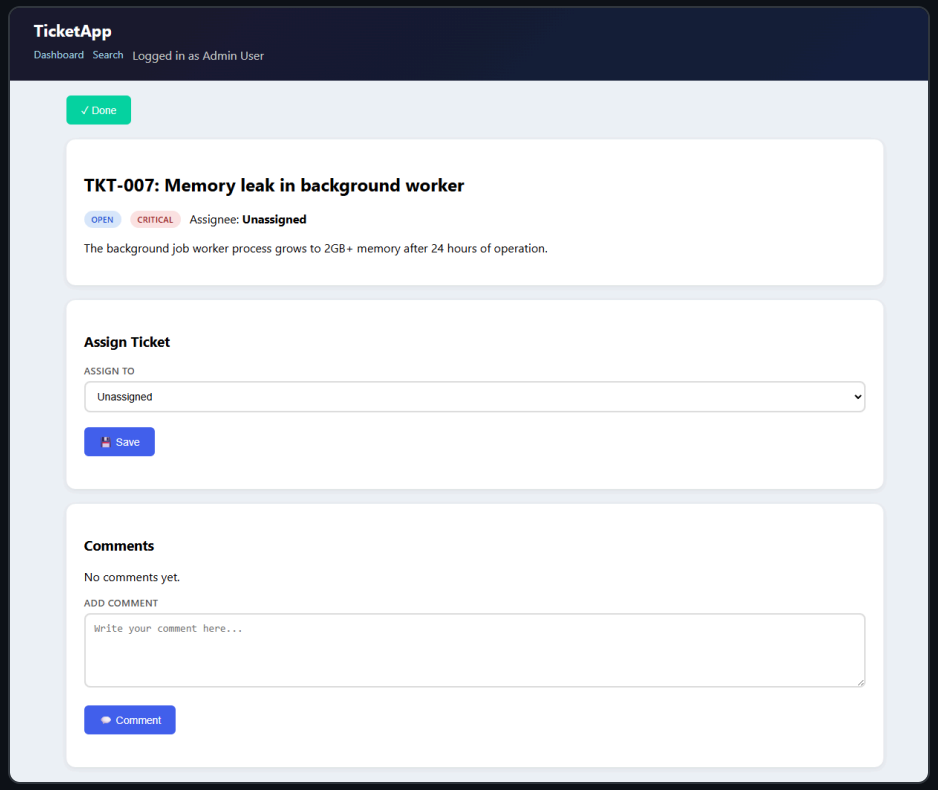
We recorded a "resolve ticket" workflow on Layout v1. Then replayed it on a **completely redesigned** Layout v2. **Zero changes to the workflow.**

Layout v1 -- Recorded Here



"Resolve" button, right-aligned, text labels

Layout v2 -- Replayed Successfully ✓



"✓ Done" icon button, top-left, different CSS

Changes handled: Button text "Resolve" → "✓ Done" | Position bottom-right → top-left | CSS classes changed | New confirmation modal auto-dismissed | Table columns reordered

Workflow Recording: Learn by Watching

TierZero learns new workflows by **watching a human do it once**. No coding required.

🎥 Record
Human demonstrates task
→
🏷️ Annotate
LLM adds meaning
→
⚙️ Generate
Intent-based workflow
→
📦 Package
Hot-loadable skill
→
▶️ Replay
Adaptive execution

What gets recorded:

- Every click, keystroke, navigation
- Page state before and after each action
- Element accessibility info (role, label)
- State changes between steps

What gets generated:

- Intent-based steps (goals, not selectors)
- Variables detected (ticket IDs, text input)
- Hot-loadable skill with manifest
- Parameterized for reuse across tickets

Key insight: The recording captures WHAT the human did. The generator converts to WHY (intent goals). Replay figures out HOW adaptively. If the UI changes, the skill still works.

Connectors: Plug Into Any System

TierZero connects to where your tickets live. Same agent brain, different ticket sources.

ServiceNow

Incidents, changes, requests. Full REST API integration. Read/write tickets, comments, assignments.

Jira

Issues, bugs, stories. Atlassian REST API. Project-scoped. Supports custom fields.

GitLab

Issues and merge requests. GitLab API v4. Labels, milestones, related MRs.

Adding a new connector: Implement the `TicketConnector` interface (7 methods: getTicket, listTickets, getComments, addComment, updateTicket, search, getEscalationTeams). TierZero handles the rest.

Event Sourcing + CQRS Core

TierZero is built on Event Sourcing. Every action the agent takes is recorded as an immutable event. Full audit trail. Time travel debugging.

Domain Aggregates:

- **Ticket** -- lifecycle events (created, commented, resolved, escalated)
- **IntentExecution** -- browser interaction events (attempted, resolved, succeeded, failed, recovered)
- **WorkflowExecution** -- end-to-end workflow events (started, step completed, finished)

Why This Matters:

- **Full audit trail** -- every decision, every action, every recovery attempt
- **Selector cache** -- successful selectors cached as read models, speeds up future runs
- **Replay debugging** -- reproduce any failure from the event stream
- **Analytics** -- resolution rates, confidence trends, escalation patterns

Skills System: Hot-Loadable Capabilities

TierZero's capabilities are modular, hot-loadable skills. Each skill provides specific domain capabilities.

🏢 ServiceNow Skill

Navigate ServiceNow, fill forms, resolve incidents, handle SSO login

🌡️ Nest Thermostat Skill

Read temperature, set schedules, eco mode

💡 Hue Lights Skill

Control Philips Hue -- turn on/off, set colors, scenes

📊 App Insights Skill

Query Azure Application Insights, investigate errors

📷 Nest Camera Skill

Check camera status, get snapshots, manage alerts

🤖 Auto-Generated Skills

Record a workflow → generate a skill. Any browser task becomes a reusable capability.

Skills are loaded from `skill.json` manifests. Drop a skill folder in, restart, done. Generated skills from workflow recording are production-ready.

The Market Opportunity

$50-400B

TAM per vertical in services automation

30-50%

RPA implementation failure rate

Why Now:

- LLMs finally good enough for real decision-making
- Vision models enable true UI understanding
- Accessibility APIs provide reliable element targeting
- Event sourcing enables enterprise-grade audit trails

Why TierZero Wins:

- **RPA:** Brittle scripts. Break on every UI change.
- **Copilots:** Suggest answers. Human still does the work.
- **TierZero:** Does the work. Autonomously. Adapts when things change.

Built in One Sunday Morning

9,605	243	18	0
lines of code	tests passing	new modules	failures

Autopilot, not copilot.

Sell outcomes, not tools.

"Your queue is zero by morning."

github.com/ellistev/tierzero